SNF-Bench++: Evaluating Fixed-Camera Video Generation Across Sustained Motion and Long Horizons

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***Abstract*—Long-running fixed-camera video generation requires stationary scene support and sustained, plausible motion in the intended region. Existing video quality and consistency measurements provide useful signals but do not by themselves establish whether generated motion has the right material, direction, or temporal evolution. This working manuscript specifies a measurement framework for text- and image-conditioned evaluation that separates support fidelity, material behavior, temporal continuity and human judgments. Its development draws on earlier SNF-Bench evaluations and a source-fixed image-to-video protocol. The independent journal confirmation set, experimental panel and human-alignment results remain to be collected; this draft makes no claim that the candidate measures have been validated.**

***Index Terms*—video generation, fixed camera, long horizon, benchmark, perceptual evaluation**

I. Introduction

In a fixed-camera video of a river, the water should move while the banks stay in place. Long generated videos can fail in either direction: the landscape may drift while the river moves, or the image may stay stable because the river has frozen. A useful evaluation must show both failures separately.

Existing benchmarks already measure important aspects of video quality, motion and consistency [2]–[6]. Our question is whether a fixed-camera evaluation adds useful information on the same videos, not whether those benchmarks are unable to see the problem. The distinction matters at long horizons, where small changes can accumulate over minutes [7]–[11].

SNF-Bench++ evaluates text-to-video and image-to-video generation for fixed-camera scenes with continuous natural motion. It separates the *background*, which should remain stable, from the *material*, such as water or smoke, that the prompt asks to move. We report background displacement and motion alongside early and late material motion. We also study checks for motion location, direction, repetition and plausibility. These checks are candidates until their controls, real-footage comparisons and human studies are complete.

The original SNF-Bench preprint [12] and the source-fixed method-paper protocol are prior work. Appendix A will distinguish their contributions from this journal study. Human and animal subjects are possible extension tracks; they are not evidence for the present nature-flow task.

This working manuscript sets out:

- **A scene suite** for both generation tracks, with planned 60, 120 and 240 second horizons and documented region maps (Section III).
- **Separate measurements** of background change, material motion and temporal behavior, with explicit limits and abstentions (Section IV).
- **A validation plan** using targeted controls, real footage, human judgments and comparisons on shared inputs (Section V).
- **An evaluation plan** for released long-video generators (Section VI). [Pending: independent panel, human validation and headline findings]

II. Related Work

Video-generation benchmarks. Multi-dimensional suites decompose quality into named dimensions and validate each against human preference [2]–[5], [13], [14]. WorldScore [6] adds source-anchored regional motion, and VMBench [15] and Video-Bench [16] target motion perception and judge models. VBench already reports a trade-off between temporal consistency and dynamic degree and keeps the two as separate dimensions [2], and WorldScore reports static and dynamic aggregates separately, measuring geometric drift by SLAM reprojection error [6]. The premise that consistency and motion must be reported separately is therefore established; what remains open in the fixed-camera, minute-scale setting is whether region-specific measures add information over these incumbents on the same videos, which Section V tests directly, including where incumbents already agree. These suites mostly evaluate clips of a few seconds and read each dimension in one direction.

Dynamics, time-lapse and physics. DEVIL [17] grades dynamics against the prompt and DIVE [18] rates image-to-video dynamics, both penalizing static outputs that would otherwise score well on consistency; ChronoMagic-Bench [19] evaluates metamorphic time-lapse content, and VideoPhy [20] and Physics-IQ [1] test physical plausibility, the latter against real continuations. None of these isolates stationary support from intended motion over minutes.

Motion and quality metrics. Distribution metrics compare generated and real videos [21], [22]. TRAJAN [23] scores single videos from point-track reconstruction. Quality-assessment datasets collect mean opinion scores for generated video [24], [25]. We evaluate such measures as comparators and candidate statistics inside role masks; their suitability for water, smoke and fire over minutes is measured, not assumed.

Long-horizon and fixed-camera generation. Autoregressive and diffusion-forcing generators extend rollouts to minutes [7]–[11], [26]. Single-image cinemagraph methods [27],

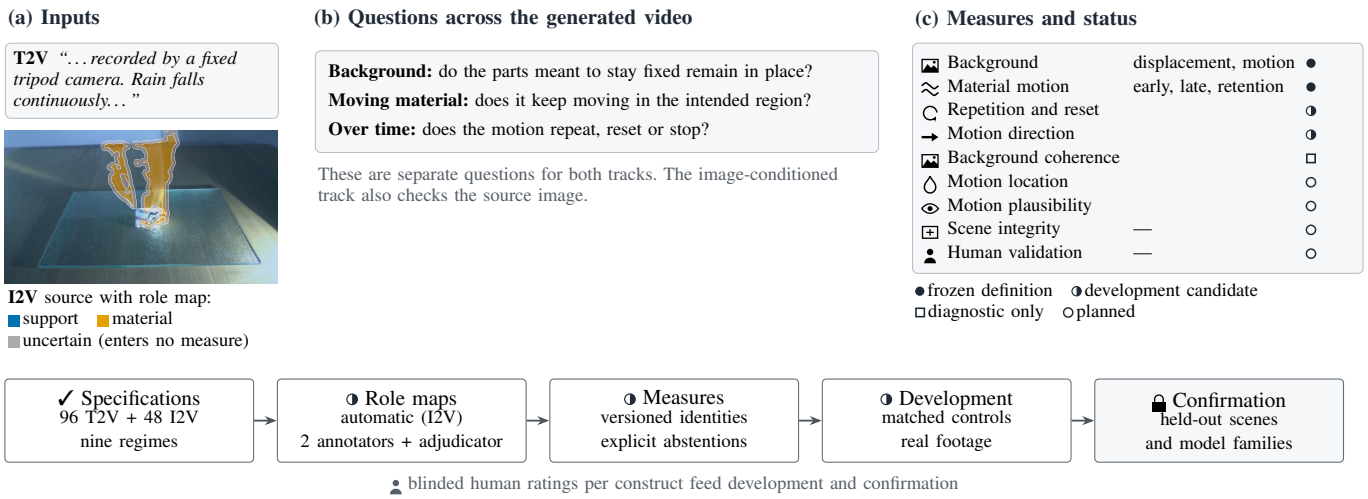


Fig. 1. **SNF-Bench++ at a glance.** (a) Each scene specifies a fixed viewpoint and names a natural process; I2V sources carry role maps; illustrated on a public fixed-camera scene, smoke rising from burning paper, with roles derived from that benchmark's motion masks (overlay pixels such as falling snow, also excluded, are absent here). (b) The questions shared by the text- and image-conditioned tracks. (c) Measures by construct and validation status; no summed score. Bottom: the protocol, from specification to held-out confirmation. Image in (a): Physics-IQ benchmark [1], scene 0140, first frame, © 2024 DeepMind Technologies Limited, CC BY 4.0; resized, with roles overlaid. An interactive version accompanies the supplementary material (`interactive/measurement_map.html`).

[28] animate a region while holding the rest still, which makes them a natural composite-type entrant.

Entrant policy. Every entrant is a published method evaluated as released. Steady-Forcing [29] is entered as its public preprint, and SceneTether [30] will be entered only once its preprint is public. Both are judged under the same rules as external methods, each as its single released configuration; no ablation or internal variant of either appears in any table. The final comparison and bibliography require verification at the journal freeze.

III. Benchmark Construction

Figure 1 summarizes the proposed workflow; its annotations, real-footage validation and held-out confirmation are still pending.

A. Task

The core task conditions on a fixed viewpoint and a specified natural process. The image-conditioned track uses a source image and prompt; the text-conditioned track has no common source image and requires its own partition and annotation policy.

B. Scene Suites and Taxonomy

Scenes are stratified by transport regime, the physical way material moves (for example channel flow, falling water, combustion and precipitation), and results are macro-averaged over regimes so that no regime dominates. Figure 2 shows the specified scene counts, not completed generations. The four water regimes together account for 42/96 text-conditioned and 21/48 image-conditioned specifications. The text-conditioned suite specifies the same scenes at every horizon; every image-conditioned source is cut to 832×480, the native resolution of the evaluated base family; any entrant whose released pipeline resizes or re-encodes is recorded as such. [Pending: caption lengths, source resolution and provenance]

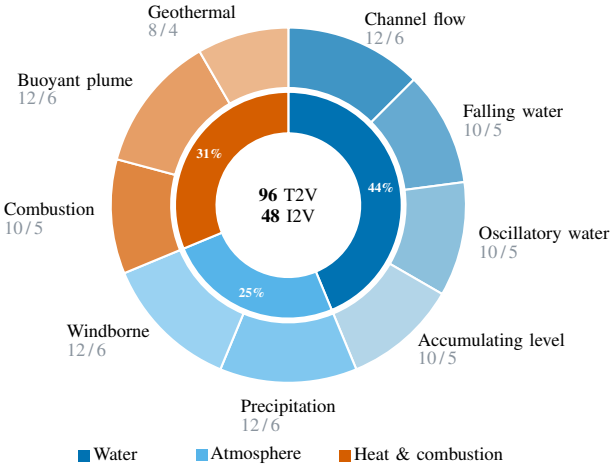


Fig. 2. **Specified scene composition.** Inner ring: medium family (display grouping); outer ring: transport regime, labelled with T2V / I2V scene counts. Each regime holds 8.3–12.5% of a track, but the four water regimes together hold 44% (42/96 and 21/48), so water is also reported as one group. Directions are declared for 69/96 T2V and 43/48 I2V scenes; they are text-derived candidates pending independent annotation. These are specifications, not completed renders, validated masks or rights clearance. Filterable list: `interactive/scene_explorer.html`.

C. Splits and Contamination Audit

Existing source sets have informed method development and therefore are not an untouched journal confirmation cohort. Each confirmation source is checked against every development, training and earlier evaluation image by exact hash, perceptual hash and geometric verification, and sources reused from earlier evaluations are reported as a separate seen-overlap

stratum. [Pending: fresh rights-cleared confirmation set and its audit counts]

D. Role Maps

The planned role maps distinguish stable support, target material, moving subjects, overlays and uncertain regions, so that, for example, a moving boat and the water it floats on can be measured separately. Observable material directions are to be annotated as image-plane sectors, with uncertain and multi-direction labels. The planned protocol uses two independent annotators, reports agreement and adjudicates or marks disagreements uncertain. In the image-conditioned track, source-derived maps supply common regions across systems; the existing maps are not yet independently human-verified. The text-conditioned track has no common layout, so its planned regions require blinded per-output annotation and uncertainty reporting. [Pending: annotation procedure, reviewer independence, agreement statistics]

E. Horizons and Generation Protocol

The planned primary horizons are 60, 120 and 240 seconds, with a short control. Same-rollout prefixes and separately requested durations are distinct experimental units. Each entrant runs with its released code, weights and default sampler. Native long-video inference, documented extensions and wrapped short-clip models are separate row types and are never pooled. Seeds are fixed per track. Each output's resolution, frame rate and encoder are recorded, and scoring resamples to a common analysis rate. Differences that remain are reported as part of the row type rather than removed. Final sampling, source rights, source-family splits, annotation procedures and model configurations must be frozen before the confirmation analysis. [Pending: entrant table with release revision, row type, seeds and compute]

IV. What We Measure

Our task asks two questions: *Does the background stay in place?* and *does the intended material keep moving?* We report the answers separately. For example, a frozen river can preserve its banks perfectly, while a moving river can be accompanied by a drifting landscape. Neither result is sufficient on its own.

The *background* is the region expected to remain stable, such as a riverbank or a building. The *material* is the region expected to move, such as water or smoke. The region maps and their limitations are described in Section III. Figure 3 summarizes how the measures use these regions.

A. Background and Motion

Table I introduces the names used in this paper. The definitions below retain the original identifiers to connect them to the published SNF-Bench definitions and stored records. A name does not change a formula, mask, reference frame, unit or version. In particular, background displacement in the earlier output-referenced protocol and the source-referenced variant must remain separately identified.

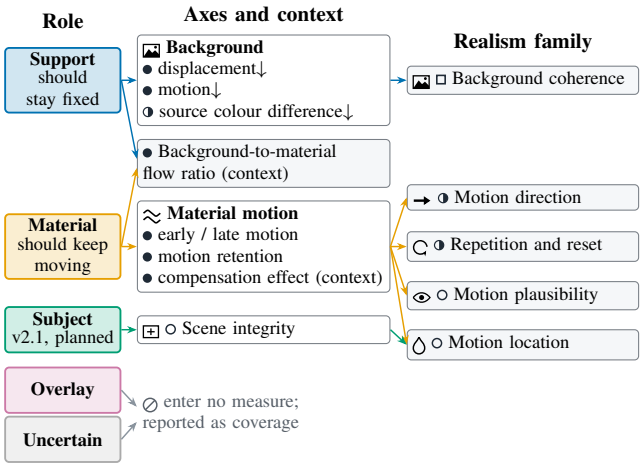


Fig. 3. **Illustrative role routing.** Lines show intended inputs, not complete computational dependencies: the background-to-material flow ratio compares roles and motion compensation can use support information. Scene integrity and subject handling are planned. Overlay and uncertain pixels are excluded from the current partitioned measures and reported as coverage. Realism measures are reported beside the axes and never summed with them. A downward arrow marks a directional support statistic, not an overall quality score; material activity and the contextual ratios require joint interpretation. ● frozen definition, construct validation pending; ◑ development candidate; □ diagnostic only; ○ planned. Interactive version: `interactive/measurement_map.html`.

TABLE I
Core measurements in plain language. These labels describe what is measured, not overall video quality. Frozen definitions remain subject to validation and eligibility checks.

Measurement	Question answered
Background displacement	How far do matched background features move relative to the chosen reference?
Background motion	How much optical-flow motion occurs in the background region?
Early and late material motion	How much material-region motion remains after estimated global motion is removed?
Motion retention	How does late material motion compare with early motion?

Background displacement (fBD) is a diagnostic: it requires trackable features and does not capture every appearance change. Background motion (NBF) can also respond to lighting changes or flow-estimation errors. Report both with the region, reference and failure counts. NBF is derived from the stored `BFR` field using the effective sampling rate; raw BFR and time-normalized NBF must not be interchanged.

Read motion retention (FP) together with the early and late material motion values (MCFF-E and MCFF-L). A retention ratio near one can arise when the material barely moves throughout, and a value above one is not automatically better. Optical flow can also be high because of flicker or an unrealistic texture. Thus neither motion amount nor retention establishes natural motion.

The historical background-to-material flow ratio (DLR) and compensation effect (DAR) are supporting diagnostics, not headline quality scores. DLR compares background flow with

raw material flow; DAR describes how estimated global-motion compensation changes the material-flow estimate. Neither is a percentage of motion caused by drift.

B. Additional Checks Under Development

The core measurements do not tell us whether motion is in the right region, follows a plausible direction, or repeats. We therefore investigate the following checks alongside them; none is included in a combined score.

- **Background coherence:** do the appearance and boundary of the stable region fit the rest of the scene? Frame-change and boundary statistics are diagnostics, not reliable detectors of pasted backgrounds.
- **Material motion location:** does the requested material move, or is the observed motion confined to another object? This requires separately verified material and subject regions.
- **Motion direction:** where a direction is observable and independently annotated, does the measured motion follow it? Ambiguous, turbulent and multi-direction cases require an appropriate label or abstention.
- **Repetition and reset:** does motion visibly replay, jump back or freeze? Natural waves and foliage can repeat, so a recurrence signal alone is not a loop verdict.
- **Motion plausibility:** does the motion resemble that of the specified material? Candidate flow, track and texture measurements will be compared with real footage and human judgments; TRAJAN, set-level FVMD and VBench-2.0 provide relevant comparisons [4], [22], [23].

These are *questions and candidate checks*, not validated realism scores. Source-matched real footage is not a unique correct continuation, and two-dimensional statistics do not prove physical correctness. The validation plan in Section V separates the failures each check should detect from natural variation it should tolerate.

When a measurement cannot be made. We report how many videos are eligible and why a measurement abstains, for example because a region is empty or registration fails. An abstention is not replaced with a score.

V. Metric Validation

The planned study asks three questions about each candidate: does it respond to the failure it is meant to detect, does it avoid flagging acceptable variation, and does it agree with people? We develop the checks on one set of scenes and model families, then assess the frozen choices on held-out data. Crops, horizons, controls and seeds from the same scene stay together. We will report uncertainty, eligibility and abstentions; synthetic controls alone cannot establish that motion looks natural.

A. Targeted Controls

Each check is tested on the failures it targets and on variations it should tolerate. We choose these tests, the comparison metrics, coverage requirements and effect thresholds before inspecting confirmation data. Controls include:

- pass-through re-encodes;
- pasted, grain-matched, feathered and illumination-matched composites;
- time-reversed, ping-pong, exact and cross-faded loops;
- freezes;
- noise warps and boiling texture;
- real footage resampled to other playback speeds;
- unrelated subject motion.

Each control is paired with an unchanged re-encode of its parent. We also test whether codec, resolution or bitrate can explain the difference. To estimate false alarms, we need independent real scenes: multiple windows from the same recording do not count as independent scenes. Even zero false alarms in 30 independent scenes leaves a one-sided 95% upper bound near 9.5%. A check that fails its tests remains a diagnostic.

B. Human Alignment

Independent source-role annotations and blinded human judgments are required. Following common practice for generated-video benchmarks [2], [3], model-level agreement is reported as the Spearman correlation between metric-derived and human-derived win ratios per dimension. Following quality-assessment practice [24], [25], video-level agreement is reported as SRCC, PLCC and KRCC against mean opinion scores. Inter-rater reliability is reported as Krippendorff's α [31]. Raters answer separate questions on support stability and integrity, material presence and location, amount of activity, motion plausibility, direction, and replay versus legitimate periodicity, each with *cannot tell* and *not applicable* options. Model-level win ratios, video-level opinion scores and paired preferences are analysed separately. A source-conditioned real-versus-generated choice is reported as a further discrimination test; because sharpness and codec cues can drive it, it does not establish physical validity. [Pending: human-study results, annotator counts and protocol]

C. Incremental Validity and Robustness

Each admitted measure is compared with the closest established measure on the same videos and controls [2], [4], [6], [22], [23]: agreement is reported, and so is any case where the new measure adds nothing. Incumbents are held to the same admission rules as the new measures, and an incumbent that passes a control is reported as passing it. Rankings are recomputed under seed, frame-sampling, encoder and mask-erosion perturbations, and their stability is reported as rank correlation with intervals. [Pending: robustness results]

VI. Benchmark Results

A. Existing Evidence

The existing source-fixed port was checked on two nine-scene, 60-second subsets against stored scores. These checks establish numerical reproduction on those inputs, not construct validity. Existing 60-second image-conditioned scores permit matched comparison of source-fixed SNF, VBench-I2V and

VBench-Long on shared source scenes. The 120- and 240-second method records score 60-second endpoint windows of separately requested generations; these should not be interpreted as trajectories of a single rollout. Historical text-to-video records use different mask and measurement identities and are reported separately.

Table II describes the available development records rather than a new leaderboard. The text-conditioned 60 s count is the unambiguous shared-identifier intersection in the existing comparison; identifiers still need video-hash verification. The 120 and 240 s counts are for the seven historical checkpoints. The image-conditioned counts describe paired examples from the previously scored family; different measures can have different eligible populations. The image-conditioned endpoint comparisons use separately requested durations, not prefixes of one long video.

Figure 4 illustrates image-conditioned outputs from four released pipelines on two sources. These are five-second examples, not the planned minute-scale comparisons. The prior 240-second text-conditioned figure reused rollout samples shown in a simultaneous Steady-Forcing submission and has been removed. [Pending: non-overlapping text-to-video qualitative comparison and longer image-to-video comparisons]

B. Leaderboard

Results are reported per track and horizon as per-axis profiles with 95% source-clustered bootstrap intervals [32], macro-averaged over regimes, with eligibility counts beside every value. No single summed score is reported. [Pending: leaderboard tables per track and horizon]

C. Per-Regime Analysis and Comparison with Established Metrics

[Pending: per-regime breakdown; agreement with VBench and WorldScore dimensions]

[Pending: rights-cleared independent confirmation set, independently reviewed role maps, native long-video baselines, sufficient multi-seed coverage, real-periodicity controls and human construct validation. Previously scored videos and metrics are reused rather than regenerated or rescored where their input and protocol identity is verified.]

VII. Findings and Discussion

[Pending: findings supported by the confirmation results, each with its interval and eligible population. Candidate questions: does motion persist or decay over the horizon; does support fidelity trade against material activity; do regimes differ in difficulty; where do established metrics disagree with the measures here.]

VIII. Limitations, Ethics and Reproducibility

Existing result populations have development exposure, and passing numerical parity does not establish perceptual validity. Source-relative geometry, activity and recurrence can each fail under plausible counterexamples. Real-reference bands depend on the real-footage set: they are reported with that set's size and medium coverage and are recomputed, not edited, when it grows.

Any final release must include exact input and model provenance, metric versions, failure and abstention counts, annotation uncertainty and rights-cleared material. Every source, real clip and generated video is released only under a recorded licence; an asset without one is withheld. The release follows a datasheet [33] (Appendix B) and versions both the inputs and the measurement identities, so later results are never pooled with earlier ones. Human-study procedures, consent and compensation are reported with the study. [Pending: maintenance plan and hosting]

Publication overlap with the original SNF-Bench [12], SceneTether [30] and Steady-Forcing [29] work must be disclosed. Metric and threshold selection do not use the standing of either author-affiliated method, and results are reported with and without those methods.

IX. Conclusion

The working protocol treats long-horizon fixed-camera video evaluation as a multi-axis measurement problem in which each statistic is read two-sided against real footage and admitted only after adversarial and human validation. Evidence for reliable naturalness assessment awaits the independent experiments and human study specified above.

Appendix A
Relation to the SNF-Bench Preprint

The SNF-Bench preprint [12] introduced region-separated drift and flow measures on an earlier prompt set. This manuscript differs in its rebuilt and contamination-audited suites, source-derived role maps, two-sided realism measures, adversarial admission protocol, human alignment and entrant panel. No earlier result is pooled with the present measurements. [Pending: itemized new-content statement with proportions, as journal policy requires]

Appendix B
Datasheet Summary

[Pending: motivation, composition, collection process, preprocessing, uses, distribution and maintenance, following [33]]

Appendix C
Additional Qualitative Comparisons

[Pending: independently selected text- and image-conditioned examples that do not reuse samples from simultaneous submissions, with complete provenance and comparable horizon coverage]

Appendix D
Known-Answer Development Results

[Pending: independent known-answer controls selected without reusing samples from simultaneous method submissions]

TABLE II

COVERAGE OF EXISTING DEVELOPMENT EVIDENCE. COUNTS ARE SCENES OR PROMPTS PER METHOD, NOT INDEPENDENT CONFIRMATION SAMPLES. TEXT- AND IMAGE-CONDITIONED PROTOCOLS USE DIFFERENT MASKS AND REFERENCES; THEIR NUMBERS SHOULD NOT BE POOLED. SOME SAMPLES ARE ALSO USED IN SIMULTANEOUS METHOD SUBMISSIONS AND ARE NOT USED AS NEW QUALITATIVE EXAMPLES HERE.

Track	60 s	120 s	240 s	Interpretation
Text to video	22 shared prompt IDs	6 prompts	4 prompts	Legacy masks; prompt sets differ across durations
Image to video	48 appearance / 41 motion	48 / 41	48 / 41	Shared source masks; 60 s endpoint windows



Fig. 4. **Image-conditioned comparison on two sources.** Source images and the final available frame of one approximately 5 s output per system. The waterfall and campfire illustrate different moving materials; each row shares the same input image and prompt across systems. The four runs use their documented inference pipelines, with configuration differences recorded in the accompanying provenance. These short examples do not establish long-horizon performance, motion realism or a ranking. Sources: VBench++ image suite [3], using Pexels images 161950 and 10462563; images are cropped and resized.

APPENDIX E
MEASUREMENT DEFINITIONS

[Pending: exact definition, eligibility rule, abstention rule and identity string of every admitted measure]

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